No.

Date.

第6次作业

C9P1

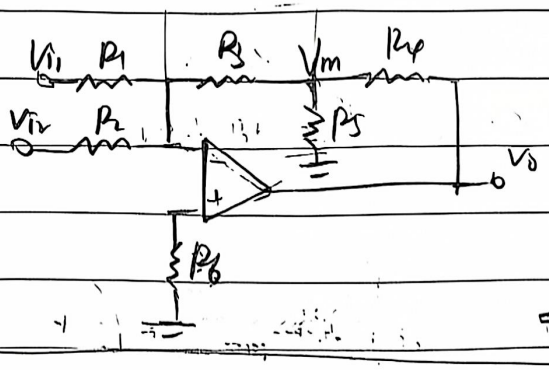
T1 D

T2 A B

C9P2

T1 D

T2

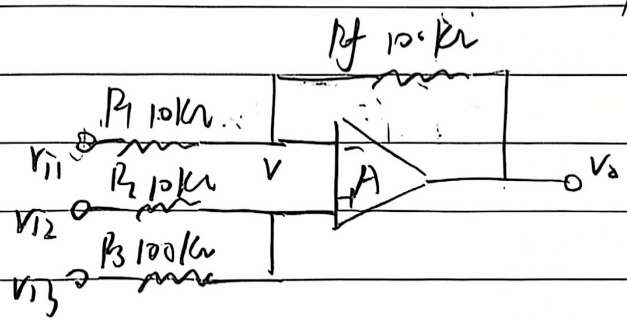


$$\begin{cases} \frac{V_{i1}}{R_1}+\frac{V_{i2}}{R_2}+\frac{V_m}{R_3}=0 \\ \frac{V_m}{R_3}+\frac{V_m}{R_5}+\frac{V_m-V_o}{R_4}=0 \end{cases}$$

$$\Rightarrow V_o=-(R_3R_4)\left(\frac{1}{R_3}+\frac{1}{R_4}+\frac{1}{R_5}\right)\left(\frac{V_{i1}}{R_1}+\frac{V_{i2}}{R_2}\right)$$

$$\left(\frac{V_{i1}}{R_1}+\frac{V_{i2}}{R_2}\right)(R_3\|R_5+R_4)=-V_o$$?

T3



$$\begin{cases} V=V_{i2}\frac{R_3}{R_2+R_3}+V_{i3}\frac{R_2}{R_2+R_3} \\ \frac{V-V_{i1}}{R_1}=\frac{V_o-V}{R_f} \end{cases}$$

$$\Rightarrow V_o=10V_{i2}+V_{i3}+10V_{i1}$$

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Date.

T4 A

T5 C

T6 $\frac{V_1-V_0}{R_1}+\frac{V_{12}-V_0}{R_2}+\frac{V_{13}-V_0}{R_3}=0$

$$\Rightarrow V_0=\frac{\frac{V_{11}}{R_1}+\frac{V_{12}}{R_2}+\frac{V_{13}}{R_3}}{\frac{1}{R_1}+\frac{1}{R_2}+\frac{1}{R_3}}$$

T7
$$\begin{cases} i=\frac{V_{11}-V_{12}}{R_1} \\ V_{01}=V_{11}+iR_2 \\ V_{02}=V_{12}-iR_2 \end{cases} \Rightarrow \begin{cases} V_{01}=V_{11}+\frac{R_2}{R_1}(V_{11}-V_{12}) \\ V_{02}=V_{12}-\frac{R_2}{R_1}(V_{11}-V_{12}) \end{cases}$$

$$\begin{cases} V=V_{02}\frac{R_4}{R_3+R_4} \\ \frac{V_{01}-V}{R_3}=\frac{V-V_0}{R_4} \end{cases}$$

$$\Rightarrow V_0=(1+\frac{R_4}{R_3})V_{02}\frac{R_4}{R_3+R_4}-\frac{R_4}{R_3}V_{01}$$

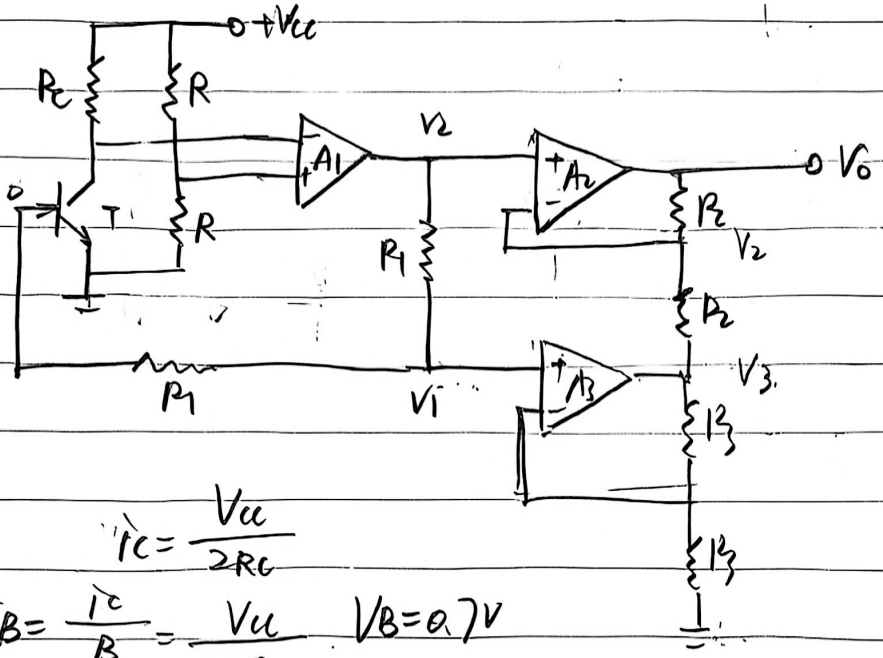
$$=\frac{R_4}{R_3}(V_{02}-V_{01})$$

$$=\frac{R_4}{R_3}(1+\frac{R_2}{R_1})(V_{12}-V_{11})$$

No.

Date.

T8



$$I_C = \frac{V_{cc}}{2R_C}$$

$$I_B = \frac{I_C}{\beta} = \frac{V_{cc}}{2\beta R_C} \quad V_B = 0.7V$$

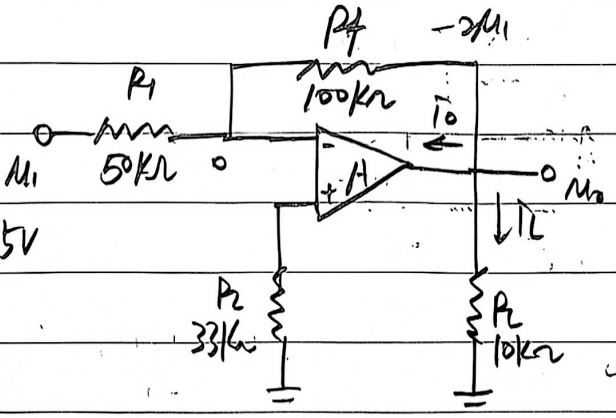
$$V_1 = V_B + I_B R_1 \quad V_2 = V_B + 2I_B R_1$$

$$V_3 = 2V_1$$

$$V_o = 2V_2 - V_3$$

$$\text{故 } V_o = \frac{V_{cc} R_1}{\beta R_C}$$

T9



V_o, I_L

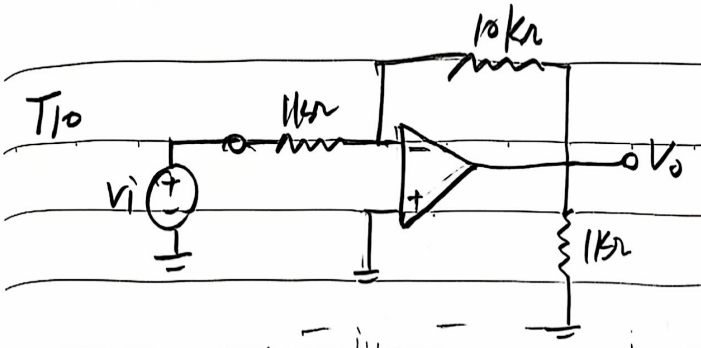
$$V_o = -2V_i = -1V$$

$$I_L = \frac{V_o}{R_L} = -0.1mA$$

No.

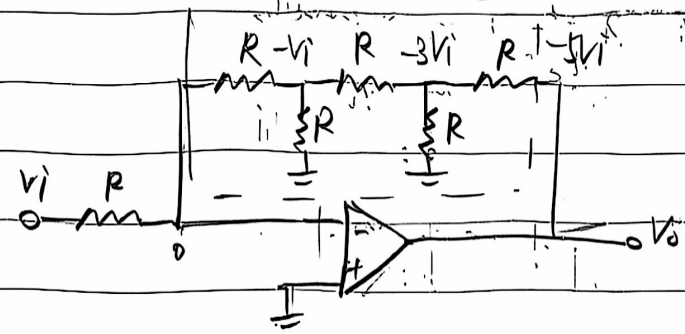
Date.

T10



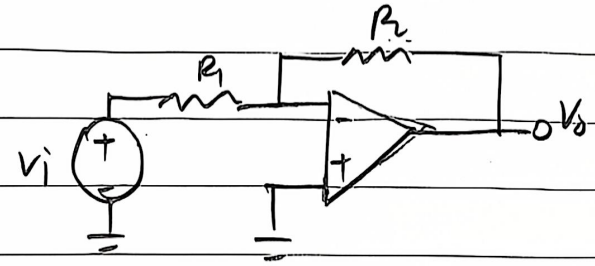
$\frac{V_o}{V_i} = 10$ 增益为 20dB

T11



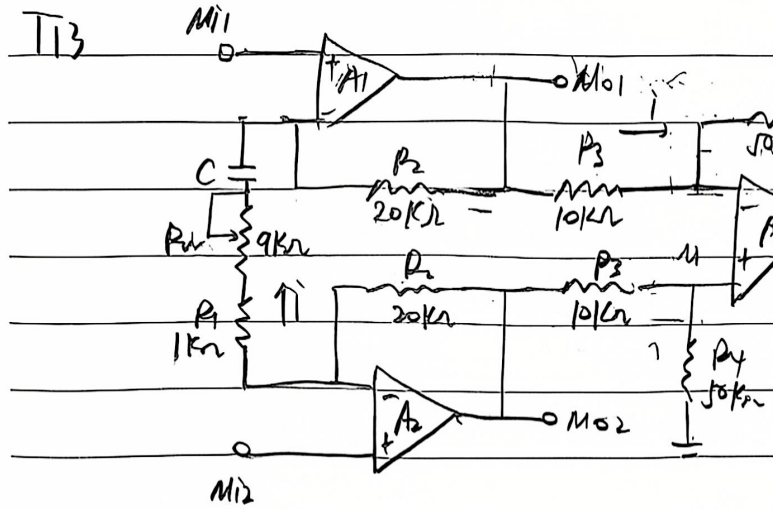
$\frac{V_o}{V_i} = -5$

T12



$$\begin{cases} \frac{R_2}{R_1} = 4 \\ R_1 + R_2 = 100 \end{cases} \Rightarrow \begin{cases} R_1 = 20k\Omega \\ R_2 = 80k\Omega \end{cases}$$

T13



$$\begin{cases} V_{o2}\frac{R_4}{R_3+R_4} = V \\ V - \frac{V_{o1}-V}{R_3}R_4 = V_o \end{cases}$$

$$V_{o1} \neq V_{i1} - \frac{V_{i2}-V_{i1}}{R_1+R_w}R_2$$

$$V_{o2} = V_{i2} + \frac{V_{i2}-V_{i1}}{R_1+R_w}R_2$$

$$\Rightarrow V_o = \frac{R_4}{R_3}(V_{o2}-V_{o1}) = \frac{R_4}{R_3}\left(1+\frac{2R_2}{R_1+R_w}\right)(V_{i2}-V_{i1})$$

故 $\frac{V_o}{V_{i2}-V_{i1}} = \frac{R_4}{R_3}\left(1+\frac{2R_2}{R_1+R_w}\right)$

$$\begin{cases} A_{vmax} = 205 \\ A_{vmin} = 21 \end{cases}$$

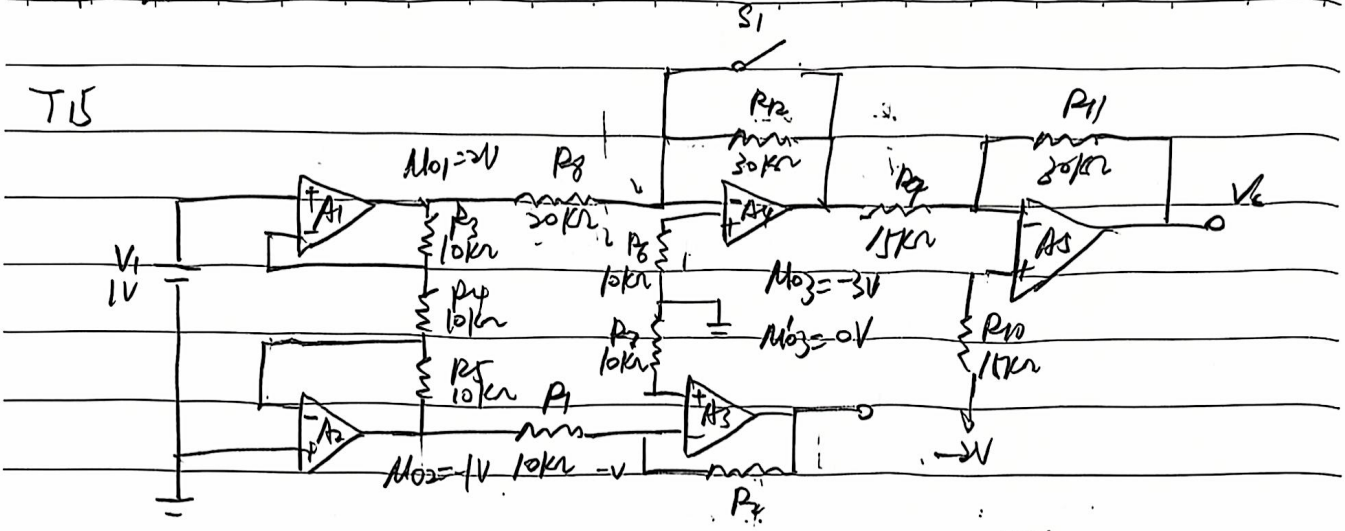
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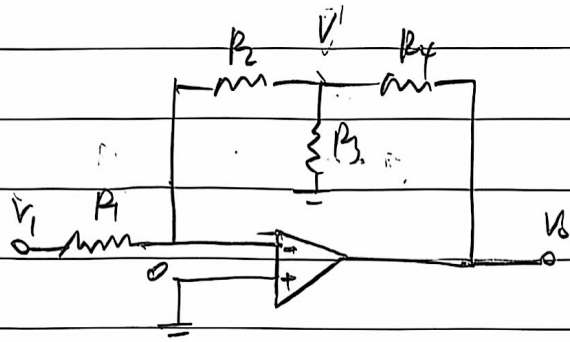
T14 A

T15



$V_o = 0V$; $V_o' = -6V$

T16



$$\begin{cases} \frac{V_i}{R_1} = \frac{-V}{R_2} \\ \frac{V}{R_2} + \frac{V}{R_3} + \frac{V - V_o}{R_f} = 0 \end{cases}$$

$$\Rightarrow V_o = -\frac{R_2 R_f}{R_1}\left(\frac{1}{R_2} + \frac{1}{R_3} + \frac{1}{R_f}\right) = -5$$

$$\Rightarrow \frac{R_3 R_f}{R_1}\left(\frac{1}{R_2} + \frac{1}{R_3} + \frac{1}{R_f}\right) = 3$$

取 $R_1 = 100K\Omega$，$R_2 = R_f = 100K\Omega$，则 $R_3 = 2.08K\Omega$

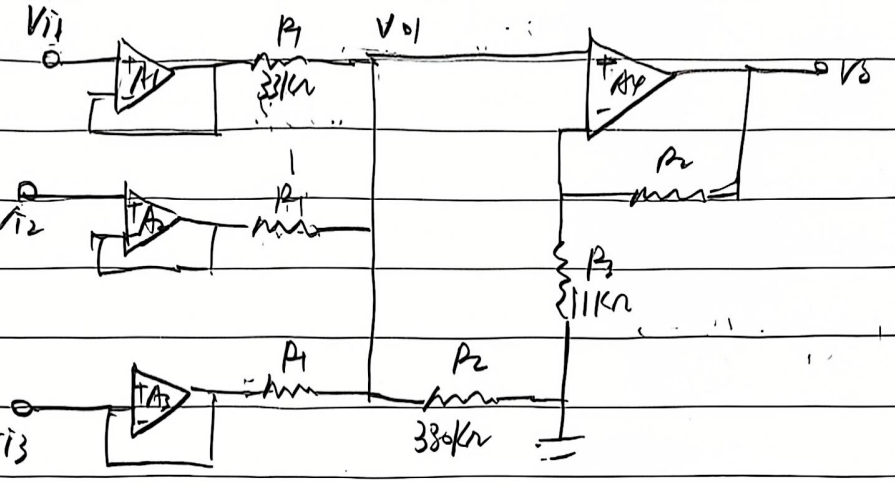
T17 C (霓加)

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Date. . .

T18 $R_1=1K\Omega$ ∞

T19



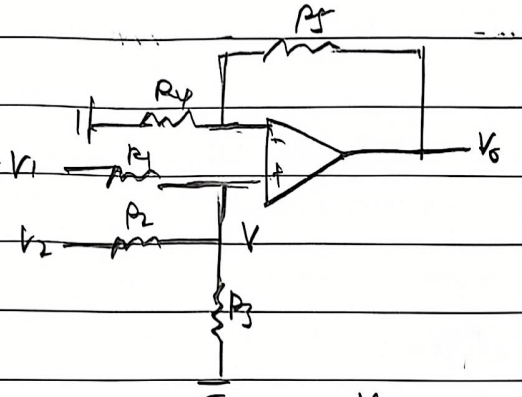
(1) $V_{o1}-\left(\frac{V_{i1}-V_{o1}}{R_1}+\frac{V_{i2}-V_{o1}}{R_1}+\frac{V_{i3}-V_{o1}}{R_1}\right)R_2=0 \Rightarrow V_{o1}=\frac{R_2}{R_1+3R_2}(V_{i1}+V_{i2}+V_{i3})$

$$V_o=\frac{R_2+R_3}{R_3}V_{o1}=\frac{R_2+R_3}{R_3}\frac{R_2}{R_1+3R_2}(V_{i1}+V_{i2}+V_{i3})$$ 10

$$\Rightarrow \frac{V_o}{V_{i1}+V_{i2}+V_{i3}}=\frac{R_2+R_3}{R_3}\frac{R_2}{R_1+3R_2}>10$$

(2) $R_0=0$

T20



可取 $R_1=1K\Omega$, $R_2=1K\Omega$, $R_3=3.3K\Omega$

$R_4=1K\Omega$, $R_f=2K\Omega$

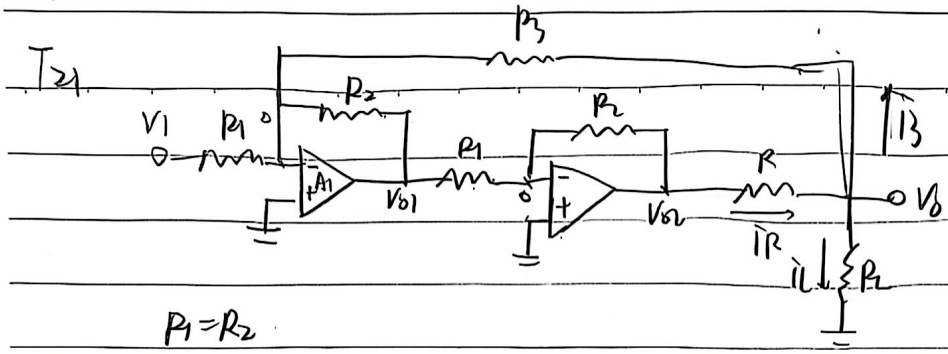
$$V_o=\left(1+\frac{R_f}{R_4}\right)\left(\frac{\frac{V_1}{R_1}+\frac{V_2}{R_2}}{\frac{1}{R_1}+\frac{1}{R_2}+\frac{1}{R_3}}\right)$$

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Date

T21



$R_1 = R_2$

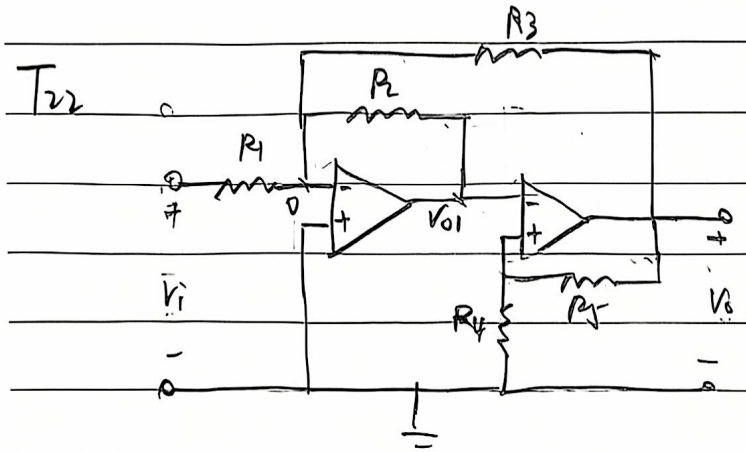
(1)

$$\begin{cases} \dfrac{V_i}{R_1} + \dfrac{V_{o1}}{R_1} + \dfrac{V_o}{R_3} = 0 \\ \dfrac{V_{o1}}{R_1} + \dfrac{V_{o2}}{R_1} = 0 \\ \dfrac{V_o - V_{o2}}{R} + \dfrac{V_o}{R_3} + \dfrac{V_o}{R_L} = 0 \end{cases} \Rightarrow V_o = \frac{V_i}{1 + \frac{R}{R_3} + \frac{R}{R_L} - \frac{R_1}{R_3}}$$

$$\Rightarrow I_L = \frac{V_i}{R + R_L\left(1 + \frac{R}{R_3} - \frac{R_1}{R_3}\right)}$$

(2) $I_L = \dfrac{V_i}{R} \Rightarrow R_3 = R_1 - R$

T22



$$\begin{cases} \dfrac{V_i}{R_1} + \dfrac{V_{o1}}{R_2} + \dfrac{V_o}{R_3} = 0 \\ \dfrac{V_{o1}}{R_4} = \dfrac{V_o}{R_4 + R_5} \end{cases} \Rightarrow \frac{V_o}{V_i} = -\frac{1}{\frac{R_1}{R_3} + \frac{R_1 R_4}{R_2(R_4 + R_5)}}$$

$$= -\frac{R_2 R_3 (R_4 + R_5)}{R_1(R_2 R_4 + R_2 R_5 + R_3 R_4)}$$

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